

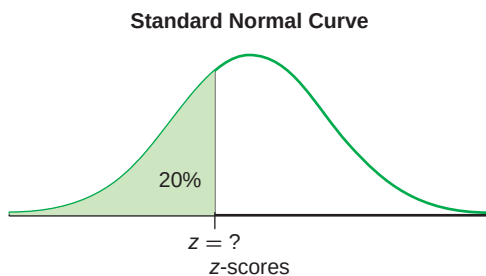
Finding Standard Scores

Areas under the standard normal curve can be used to find standard scores and percentiles.

Using Area to Find a z-Score

Although the straightforward way of reading the z -distribution table is to look up the z -score and then find the corresponding area under the standard normal curve, we can also read the table in the other direction.

Suppose that about 20% of the data in a data set are below a certain value. What z -score corresponds to this value?



► **Remember** z -scores are sometimes called standard scores.

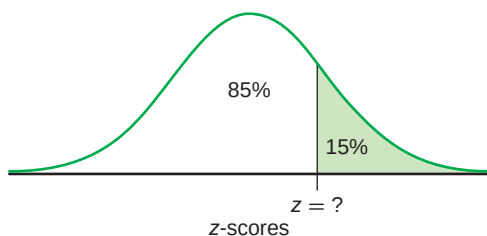
Since $20\% = 0.2$, you want to find an area in the body of the z -distribution table that is closest to 0.2.

<i>z</i>	.00	.01	.02	.03	.04	.05
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266

The area 0.2005 is closest to 20%. Use the row and column of the area to locate the corresponding z -score, which corresponds to a z -score of -0.84 .

Determining Area Above a z -Score

Suppose that 15% of the data in a data set are above a certain value. What z -score corresponds to this value?



Since the z -distribution table only gives the areas that are below a particular z -score, you need to find the percent of data below the value: $100\% - 15\% = 85\%$. Since $85\% = 0.85$, you want to find an area in the body of the z -distribution table that is closest to 0.85.

<i>z</i>	.00	.01	.02	.03	.04	.05
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749

The area 0.8508 is closest to 85%, which corresponds to a z -score of 1.04.

Understanding Percentiles

A **percentile rank** is the percentage of data that falls below a particular value. For example, if a student scores higher than 60% of the other students who took an exam, then the student's percentile rank is at the 60th percentile.

► **Remember** The n th percentile is the value (or score) below which n percent of the data may be found.

Scores on the math section of a standardized aptitude test are normally distributed with a mean of about 500 and a standard deviation of about 100. According to this information, what score would be at the 80th percentile?

Using the z -distribution table, you see that the area 0.8023 is closest to 80%, which corresponds to a z -score of 0.84. Use the appropriate formula to find the corresponding score.

$$\begin{aligned}x &= z \cdot \sigma + \mu \\&= 0.84 \cdot 100 + 500 \\&= 584\end{aligned}$$

So a score of 584 on the math section of this standardized test is at the 80th percentile.